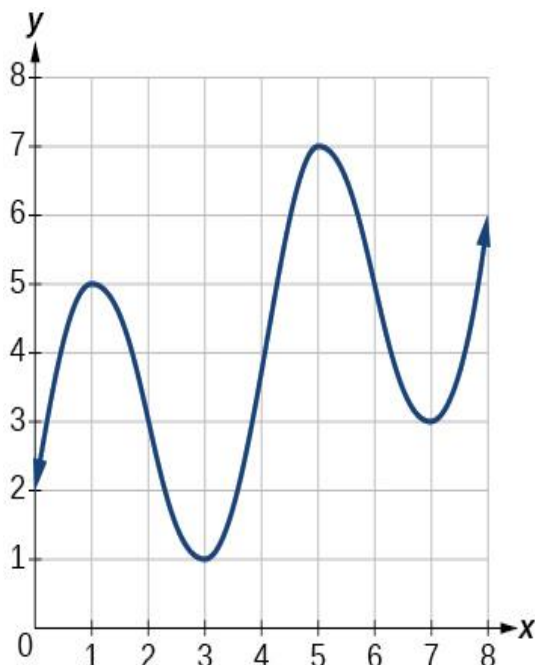


Section 1.3 Exercises

Directions: For the exercises 1 - 11, find the average rate of change of each function on the interval specified for real numbers b or h .

1. $f(x) = 4x^2 - 7$ on $[1, b]$
2. $g(x) = 2x^2 - 9$ on $[4, b]$
3. $p(x) = 3x + 4$ on $[2, 2 + h]$
4. $k(x) = 4x - 2$ on $[3, 3 + h]$
5. $f(x) = 2x^2 + 1$ on $[x, x + h]$
6. $g(x) = 3x^2 - 2$ on $[x, x + h]$
7. $a(t) = \frac{1}{t+4}$ on $[9, 9 + h]$
8. $b(x) = \frac{1}{x+3}$ on $[1, 1 + h]$
9. $j(x) = 3x^3$ on $[1, 1 + h]$
10. $r(t) = 4t^3$ on $[2, 2 + h]$
11. $f(x) = 2x^2 - 3x$ on $[x, x + h]$

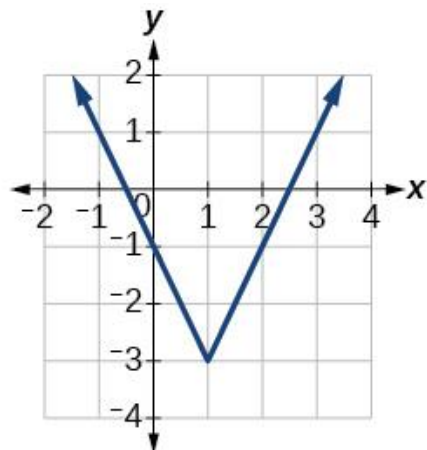
Directions: For the exercises 12 - 13, consider the graph of f shown in the graph below.



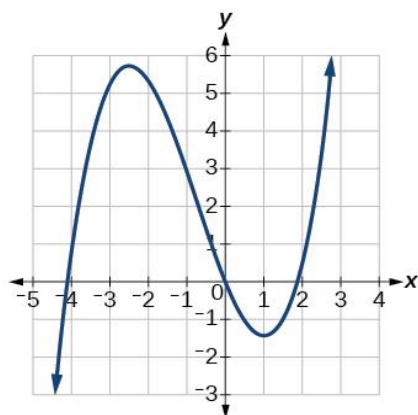
12. Estimate the average rate of change from $x = 1$ to $x = 4$.
13. Estimate the average rate of change from $x = 2$ to $x = 5$.

Directions: For the exercises 14 - 17, use the graph of each function to estimate the intervals on which the function is increasing or decreasing.

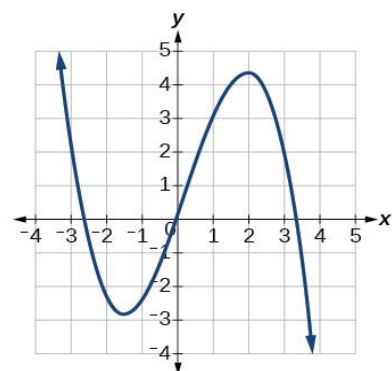
14.



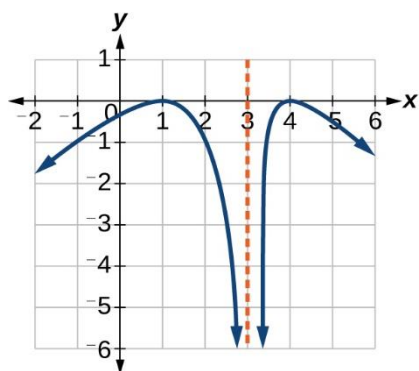
15.



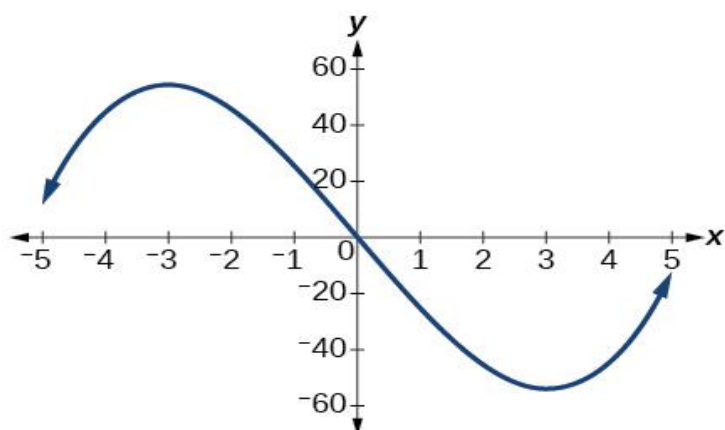
16.



17.

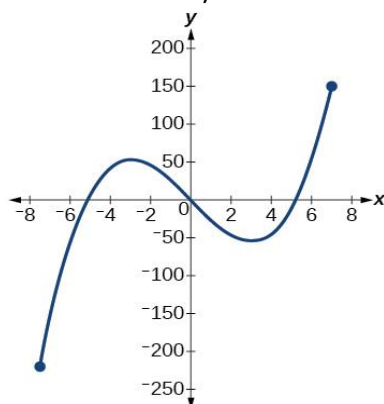


Directions: For the exercises 18 - 19, consider the graph shown below.



18. Estimate the intervals where the function is increasing, decreasing, and constant.
19. Estimate the point(s) at which the graph of f has a local maximum and a local minimum.

Directions: For the exercises 20 - 21, consider the graph shown below. Assume the complete graph is shown.



20. Estimate the intervals where the function is increasing, decreasing, and constant.
21. Estimate the point(s) at which the graph has an absolute maximum and absolute minimum.

Directions: For exercises 22 - 28, find the average rate of change of each function on the interval specified.

22. $f(x) = x^2$ on $[1, 5]$

23. $h(x) = 5 - 2x^2$ on $[-2, 4]$

24. $q(x) = x^3$ on $[-4, 2]$

25. $g(x) = 3x^3 - 1$ on $[-3, 3]$

26. $y = \frac{1}{x}$ on $[1, 3]$

27. $p(t) = \frac{(t^2-4)(t+1)}{t^2+3}$ on $[-3, 1]$

28. $k(t) = 6t^2 + \frac{4}{t^3}$ on $[-1, 3]$